



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO AMUTHA SIVAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

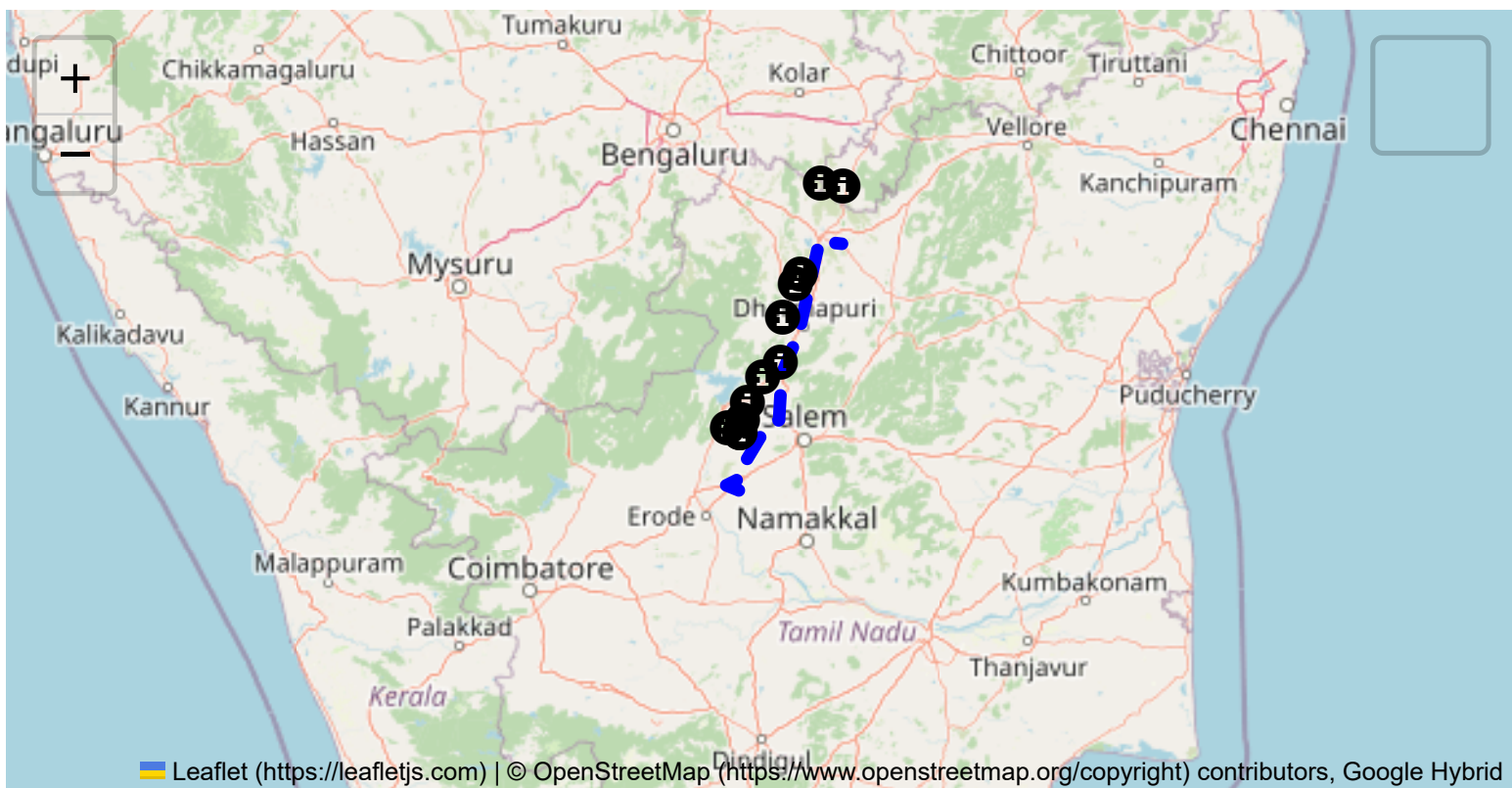
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

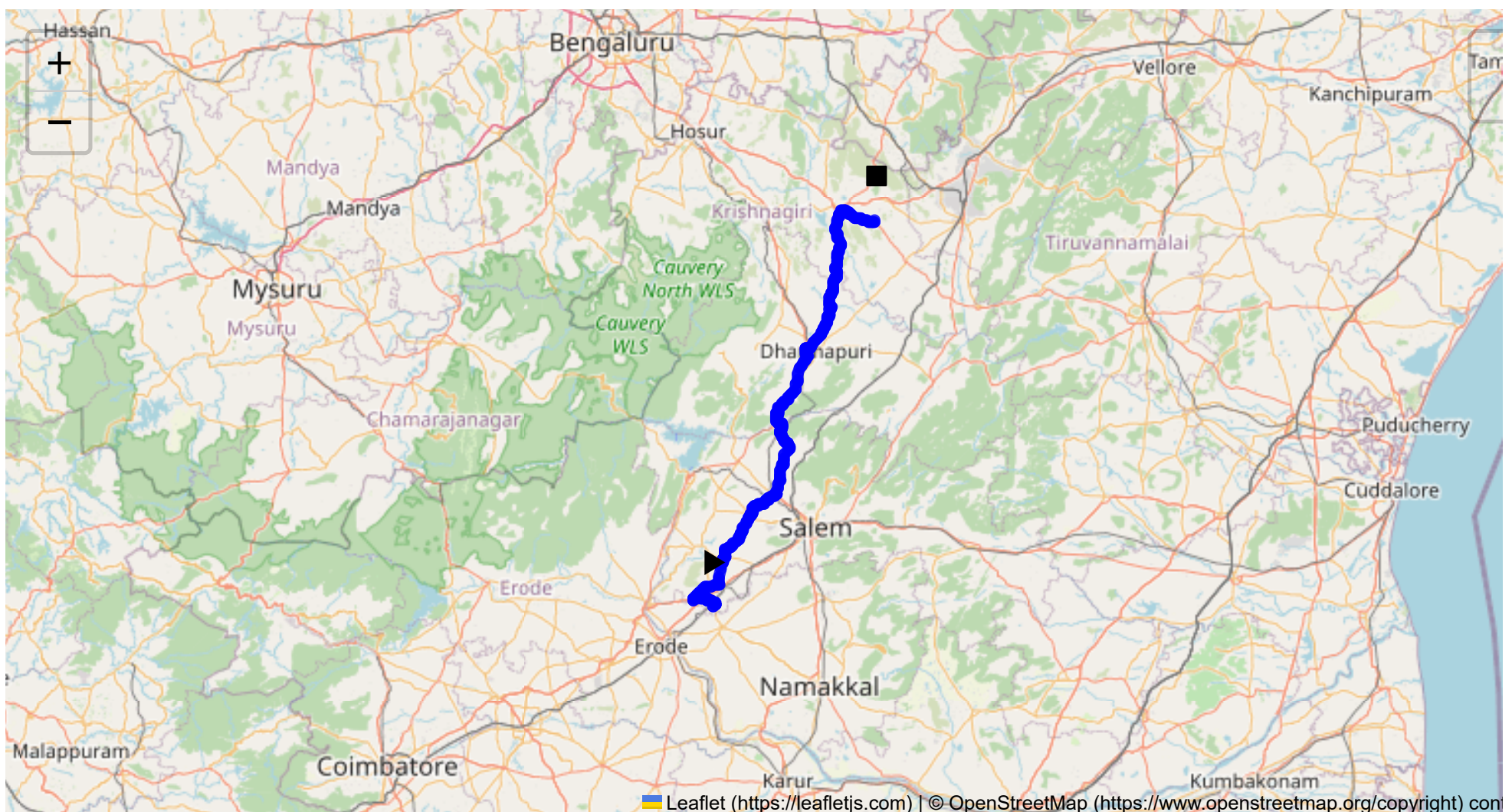
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 163.33 km
Estimated Duration: 3.2 hours
Adjusted Duration (Heavy Vehicle): 3.9 hours
Start: (11.4381, 77.8734)
End: (12.489564, 78.322956)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans a distance of approximately 163.33 kilometers from Sangagiri to Jagadevi in Tamil Nadu. It travels through several towns and major highways, including Salem-Kochi Highway, Omalur-Sankagiri Road, and Mettur Main Road. It passes through various districts like Salem, Dharmapuri, and Krishnagiri, connecting a series of rural and semi-urban areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with moderate rainfall primarily during the northeast monsoon (October to December). Hence, there could be risks of waterlogging and reduced visibility during these months. Summers (March to June) can be extremely hot, which may affect vehicle performance and driver comfort.

3. Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Traffic is generally heavier in the morning (7-10 AM) and evening (5-8 PM) due to local commute patterns.
- **Congestion Zones:** Urban sections such as Salem and Dharmapuri experience higher traffic density, causing potential delays. Key junctions, market areas, and near schools can also be congestion-prone.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Varies from well-maintained highways to narrower rural roads with potential potholes.
- **Infrastructure:** Important bridges and flyovers usually found in good condition, but caution is advised on smaller village roads that might lack street lighting.

5. Suggestions for Alternative Routes for Emergencies

- Consider bypassing major towns during heavy traffic.
- Utilize the Salem-Kochi Highway for a more direct route to avoid congested city centers.
- It's advisable to plan detours based on real-time traffic updates, particularly using state roads SH 86 and NH 44.

6. Local Regulations Affecting Hazardous Material Transport

- Ensure compliance with Tamil Nadu transport regulations, including specific permits for hazardous materials.
- Transport during nighttime might be restricted in certain zones due to safety and local regulations.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been incidents on highways involving heavy vehicles, often due to brake failure or driver fatigue. Specific data on hazardous materials may not be detailed, but common risks include spills or leaks leading to traffic disturbances.

8. Environmental Considerations and Sensitive Areas

- Proximity to water bodies in Chennai and rural agricultural zones necessitates careful driving to avoid spills.
- Avoid littering or contamination near agricultural fields and water sources.

9. Analysis of Communication Coverage

Coverage is generally good except in some rural pockets where dead zones might exist. It’s ideal to ensure the availability of portable communication devices or satellite phones for critical communications.

10. Estimated Emergency Response Times

- **Urban Areas:** Response is typically quicker due to proximity to hospitals and fire stations, often within 30 minutes.
- **Rural or Remote Areas:** Response times could extend up to 60-90 minutes based on accessibility and road conditions.

11. Summary of Risk Assessment

Overall, the route is moderately safe but requires caution due to variable road conditions, traffic densities, and weather-related risks. It's crucial for transporters to be prepared for emergencies and have alternative routes planned. Adhering strictly to safety and regulatory guidelines for hazardous materials is vital to minimizing risks.

Conclusion

The journey from Sangagiri to Jagadevi in Tamil Nadu involves navigating various terrains and urban-rural landscapes. For a safe trip, truck drivers must remain vigilant about road conditions, weather patterns, traffic densities, and compliance with regulations governing hazardous materials. Preparedness for emergencies and alternative routing is essential for efficient and safe transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	Medium	11.45545, 77.81812	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
7	Turn	High	11.45584, 77.81695	15 KM/Hr
8	Turn	Medium	11.45656, 77.81709	30 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56352, 77.90078	30 KM/Hr
11	Turn	Medium	11.59189, 77.90645	30 KM/Hr
12	Turn	Medium	11.59328, 77.90809	30 KM/Hr
13	Turn	Medium	11.63867, 77.94841	30 KM/Hr
14	Turn	High	11.64217, 77.95200	15 KM/Hr
15	Turn	Medium	11.64712, 77.95118	30 KM/Hr
16	Turn	Medium	11.67858, 77.96834	30 KM/Hr
17	Turn	High	11.70233, 77.97869	15 KM/Hr
18	Turn	Medium	11.71394, 77.99572	30 KM/Hr
19	Turn	Medium	11.71916, 78.01117	30 KM/Hr
20	Turn	Medium	11.72504, 78.01680	30 KM/Hr
21	Turn	Medium	11.73766, 78.03445	30 KM/Hr
22	Turn	High	11.73651, 78.03868	15 KM/Hr
23	Turn	Medium	11.74257, 78.04708	30 KM/Hr
24	Turn	Medium	11.86805, 78.08353	30 KM/Hr
25	Turn	Medium	11.87913, 78.08206	30 KM/Hr
26	Turn	Medium	11.92566, 78.06503	30 KM/Hr
27	Turn	Medium	11.93307, 78.06426	30 KM/Hr
28	Turn	Medium	11.93545, 78.06130	30 KM/Hr
29	Turn	Medium	11.93694, 78.05425	30 KM/Hr
30	Turn	Medium	11.95531, 78.04698	30 KM/Hr
31	Turn	High	11.95803, 78.04964	15 KM/Hr
32	Turn	Medium	11.96068, 78.04960	30 KM/Hr
33	Turn	High	11.96370, 78.05142	15 KM/Hr
34	Turn	High	11.96542, 78.04938	15 KM/Hr
35	Turn	Medium	11.96650, 78.04927	30 KM/Hr
36	Turn	Medium	11.96890, 78.05109	30 KM/Hr
37	Turn	Medium	12.12094, 78.13269	30 KM/Hr
38	Turn	Medium	12.14025, 78.12819	30 KM/Hr
39	Turn	High	12.14099, 78.12862	15 KM/Hr
40	Turn	High	12.14056, 78.13957	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
41	Turn	High	12.24611, 78.19519	15 KM/Hr
42	Turn	Medium	12.25200, 78.20034	30 KM/Hr
43	Turn	High	12.40026, 78.21775	15 KM/Hr
44	Turn	Medium	12.40734, 78.22344	30 KM/Hr
45	Turn	High	12.50709, 78.22230	15 KM/Hr
46	Turn	High	12.51556, 78.23049	15 KM/Hr
47	Turn	Medium	12.51540, 78.24044	30 KM/Hr
48	Turn	Medium	12.48362, 78.31717	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
29	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
33	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
30	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
31	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
32	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45545, 77.81812



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56352, 77.90078



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59189, 77.90645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59328, 77.90809



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71916, 78.01117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72504, 78.01680



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73766, 78.03445



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74257, 78.04708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86805, 78.08353



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87913, 78.08206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92566, 78.06503



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93307, 78.06426



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93545, 78.06130



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93694, 78.05425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95803, 78.04964



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96068, 78.04960



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96890, 78.05109



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.14025, 78.12819



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14099, 78.12862



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14056, 78.13957



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.24611, 78.19519



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25200, 78.20034



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.40026, 78.21775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40734, 78.22344



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.50709, 78.22230



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51540, 78.24044



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.48362, 78.31717
